

Q#1 : [10.5 points] fill the table below with the

1	2	3	4	5	6	7

- The main difference between a register and a counter is _____.
 a) A register has no specific sequence of states
 b) A counter has no specific sequence of states
 c) A register has capability to store one bit of information but counter has n-bit
 d) A register counts data
- In serial shifting method, data shifting occurs _____.
 a) One bit at a time
 b) simultaneously
 c) Two bit at a time
 d) Four bit at a time
- How can parallel data be taken out of a shift register simultaneously?
 a) Use the Q output of the first FF
 b) Use the Q output of the last FF
 c) Tie all of the Q outputs together
 d) Use the Q output of all FF
- The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register with an initial state 01110. After three clock pulses, the register contains _____.
 a) 01110
 b) 00001
 c) 00101
 d) 00110
- Assume that a 4-bit serial in/serial out shift register is initially clear. We wish to store 1100. What will be the 4-bit in register after the second clock pulse? (left-most bit first)
 a) 1100
 b) 0011
 c) 0000
 d) 1111
- If two inputs are active on a priority encoder, which will be coded on the output?
 a) The higher value
 b) The lower value
 c) Neither of the inputs
 d) Both of the inputs
- Which gate is added to the inputs to convert OR gate into NAND gate? _____.
 a) XOR
 b) AND
 c) OR
 d) NOT

Q#2 [5 points]: Draw the state diagram of sequence detector, which detects a pattern of three or more consecutive 0's in a string of bits coming through an input line.

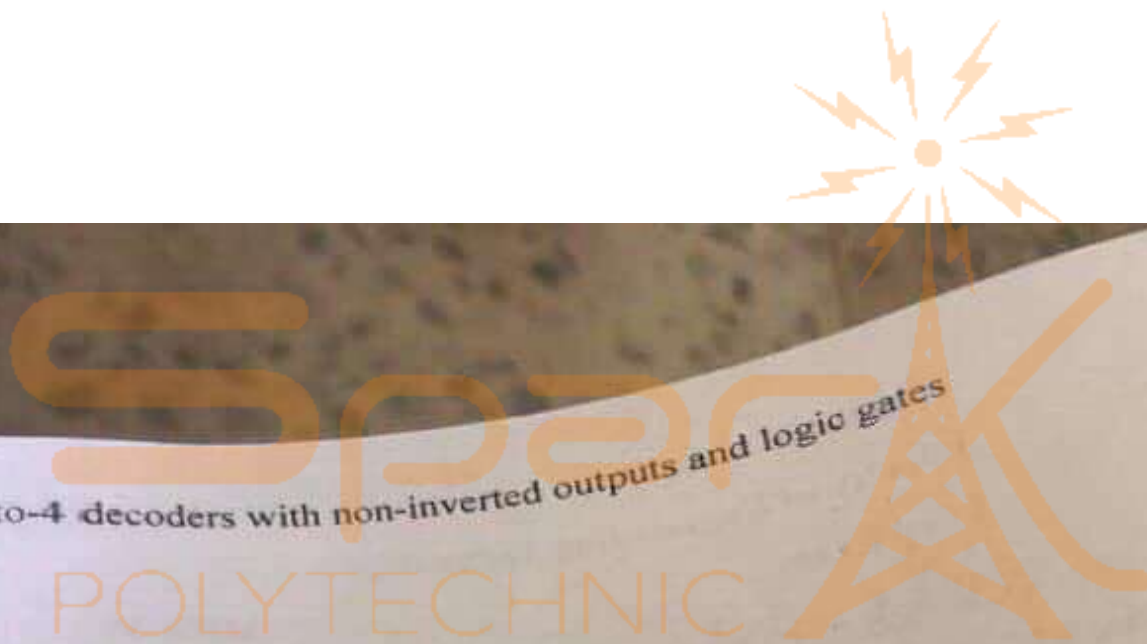
7. Which gate is added to the inputs to convert OR gate into NAND gate?
a) XOR b) AND c) OR d) NOT

Q#2 [5 points]: Draw the state diagram of sequence detector, which detects a pattern of three or more consecutive 0's in a string of bits coming through an input line.



Q#3 :| 8.5points|Implement the equation $F(A,B,C,D) = \Sigma (0,4,5,6,7,10,11,12)$ using:

1. 4-to-1 multiplexers and logic gates.



2. 2-to-4 decoders with non-inverted outputs and logic gates

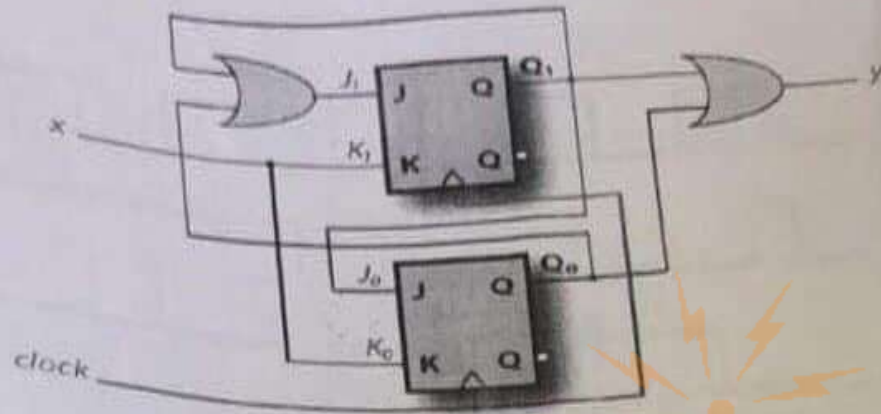
POLYTECHNIC

Q#4: [4 points] Draw the circuit of 2 bit universal register

Selection line		action
S1	S0	
0	0	Shift left
0	1	Parallel load
1	0	Shift right
1	1	No change (hold data)



Q#5: [8 points]: Consider the circuit shown below



1. Derive the equations

$J_1 =$

$K_1 =$

$J_0 =$

$K_0 =$

$Q_1(t+1) = Q_0(t+1) =$

2. Fill the table below

Present state		Input		Next state				output
Q_1	Q_0	X		J_1	K_1	J_0	K_0	Y

3. Draw state diagram



Answer one question only from Q#6 and Q#7

Q #6[4 points] Draw the (Q) for the JK flip-flop with Positive Edge Triggering Clock.

